#8 · 5.9% of marks

Ch 12 — Interest Rate Risk Management

Numerical every attempt, different tool each time: FRA (Sep 2025), cap & floor (Jan 2026), gap / Earnings at Risk + zero-cost collar (May 2026). SM TYK theory Qs: **Forward Rate Agreements** · **swaptions and their uses**.

① Interest rates & benchmarks (SM 1–2)

→ *Interest = cost of borrowing + compensation for service and **risk** of lending (default) + protection against **inflation**.*
__Factors__ (macro): (a) demand / supply of money (b) inflation (c) government — biggest borrower; RBI via note issue / __OMO__ changes CRR, SLR, bank rate

→ ***Benchmark (reference) rate**: basis for other rates — loans, overdrafts, mortgages, forwards, futures, options and especially **swaps**; floating loans = benchmark + bps by credit rating; decided by an independent body; also used to value bonds*

→ ***LIBOR**: manipulation news 2012 → decided 2017 to end → from **1 Jan 2022** use **Alternative Reference Rates (ARRs)**. ARRs differ: (i) based on **actual overnight transactions** (secured or unsecured), LIBOR unsecured and relied on panel banks' judgment (ii) **near risk-free, no term premium***

Region	ARR	Regulator	Nature
USA	**SOFR** — Secured Overnight Financing Rate	Federal Reserve Bank of New York	Secured
UK	**SONIA** — Sterling Overnight Index Average	Bank of England	Unsecured
Europe	**€STER** — Euro Short-Term Rate	European Central Bank	Unsecured
Japan	**TONAR** — Tokyo Overnight Average Rate	Bank of Japan	Unsecured
Switzerland	**SARON** — Swiss Average Rate Overnight	SIX Swiss Exchange	Secured

***India**: repo, PLR, MCLR, but most common are **MIBOR** (rate bank charges a borrower) and **MIBID** (rate at which bank would borrow from another bank) — used in IRS, FRA, floating-rate debentures.*

Secured ARRs = SOFR and SARON (both start with S). The other three are unsecured.

② Interest rate risk (SM 3)

*Change in bond prices from rate changes + impact on interest income → **price and reinvestment** risks both matter.*

THEORY + GAP NUMERICAL

③ Types of interest rate risk (SM 3.1)

Type	SM meaning
Gap / mismatch	Assets, liabilities, off-BS items with different principal, maturity or **re-pricing dates**. **Positive gap** = RSA > RSL → rate **rise raises NII**. **Negative gap** = RSL > RSA → NII rises when rates **fall**. **Earnings at Risk (EaR) = gap × assumed rate change**; bank fixes an EaR limit (based on last / current income) and a trigger for on- / off-BS hedging.
Basis	Rates on different assets / liabilities change by different magnitudes (eg assets fixed, liabilities floating); high for banks funding composite assets from composite liabilities; NII expands = favourable basis shift
Embedded option	Prepayment of loans, call / put on bonds, premature withdrawal of deposits; faster and bigger rate moves → bigger risk; adjust gap statements, charge penalties based on opportunity cost
Yield curve	Yield-curve movements over business cycles affect portfolio values and income
Price	Assets sold before maturity; prices and yields inverse; tied to **trading book** → limit size, holding period, duration, defeasance period, stop-loss, MTM
Reinvestment	Uncertainty about rate at which future cash flows can be reinvested; mismatches vary NII

Gap report — 5 limitations (SM)

(1) only timing difference of re-pricing; ignores **basis and embedded option** risk (2) assumes all items in a time band re-price together; ignores effect on **economic / market value** (3) ignores timing differences in payments caused by rate changes (4) assumes **parallel shift** of yield curve — seldom happens (5) ignores variability in **non-interest** revenue & expenses

④ Measuring interest rate risk (SM 3.2)

- → Must identify and quantify first; system should cover gap, basis, embedded option, yield curve, price, reinvestment and net interest position risks
- → Techniques: **maturity gap analysis** (sensitivity of **earnings**) · **duration** (sensitivity of **capital**) · **simulation** · **VaR** — often used in combination / hybrid

May 2026 Q5(b) (ICAI): re-pricing gap = RSA − RSL (negative); EaR = gap × rate change × ¼ (quarter) vs limit of 8% of NII → hedge if EaR exceeds the limit. Rates up hurts a negative gap; rates down helps.

5 Hedging methods (SM 4)

Traditional: (i) Asset Liability Management (ALM) (ii) Forward Rate Agreement (FRA). **Modern**: (i) Interest Rate Futures (ii) Interest Rate Options (iii) Interest Rate Swaps (+ swaptions).

6 ALM (SM 4.1.1)

Evolved by **RBI**; introduced in Indian banks w.e.f. **1 April 1999**. Comprehensive, dynamic framework to measure, monitor and manage **liquidity, interest rate, forex, equity and commodity price** risks, integrated with business strategy; structure the balance sheet so **net interest earnings are maximised** within risk preference. Extends to liquidity, market and trading risk, funding & capital planning, profit planning, growth projection. First step in long-term strategic planning — an **intermediate-term planning** function.

7 Forward Rate Agreement (SM 4.1.2) — TYK Q1

- → Borrower / lender protects against adverse rate changes; **OTC** (not exchange-traded); used by banks to fix cost of future deposits / revenue of variable loans indexed to SOFR, €STER, MIBOR
- → **Off-balance-sheet**; **no transfer of principal** (notional); **cash-settled** for the difference
- → RR > FR → **seller pays buyer** (buyer = borrower protected); RR < FR → buyer pays seller
- → Settled at the **start** of the period → difference **discounted at the actual (RR) rate**

Settlement = N × (RR − FR) × (dtm/DY) ÷ [1 + RR × (dtm/DY)] (DY = 360 or 365)

SM eg: **1 × 4** FRA (deposit placed after 1 month, matures 4 months from contract) at 5%, $1 m, 90 days; SOFR 6% → 1 bp = $25 → $2,500 ÷ [1 + 0.06 × 90/360] = **$2,463.05** to buyer.

TYK 1 (app-computed): ₹60 cr, 3 months in 6 months, FRA 9.30% → at 9.60% company **receives ₹4,39,453**; at 8.80% company **pays ₹7,33,855**. TYK 2: 6 × 9 FRA ₹100 cr at 5.25%, 3-month MIBOR 5.50% → **+25 bp**, settlement ≈ **₹6,16,523**.

8 Interest rate futures (SM 4.2.1)

- → Futures on an **interest-bearing asset** — bonds are the underlying, not the rate. Borrower fearing a **rise sells** futures (prices fall when rates rise → profit on buying back)
- → NSE: futures on **6, 10 and 13-year GoI securities** and **91-day T-bill**. Settlement: **daily MTM** + **physical delivery** any day in expiry month; final settlement only on expiry date
- → **Conversion factor** (published by NSE; introduced by RBI): CF × futures price = delivery price of a given bond
- → **Cheapest to deliver**: seller's profit = futures price × CF − quoted spot price; CTD = bond that **maximises profit / minimises loss**

NUMERICAL CORE

⑨ Interest rate options (SM 4.2.2)

Also called **Interest Rate Guarantee (IRG)**: a right, not obligation — insurance against adverse moves while keeping favourable ones. An IRO is basically a **series of FRAs** exercisable at set rates each period.

Option	Who / what	Payoff (SM)
Cap (call)	Buyer (**borrower**) pays premium; receives difference when benchmark > cap rate; no obligation if below. Evolved from IRGs on floating-rate loans (1985); LBO boom — lenders required caps. Features: notional, benchmark, cap rate (= strike), period with reset / payment dates	**$N \times \max(0, r - r_c) \times d_t/\text{days}$**
Floor (put)	Buyer (**lender / depositor**) protected from **fall** in rates; = right to **sell an FRA** → European put on an FRA; multi-period floor = bundle of puts on a sequence of FRAs	**$N \times \max(0, r_f - r) \times d_t/\text{days}$**
Collar	**Buy a cap + sell a floor** → rate bounded on both sides; cheaper than a cap alone (floor premium offsets); **zero-cost collar** = floor premium exactly offsets cap premium; gives up benefit of a big fall	**$N \times [\max(0, r - r_c) - \max(0, r_f - r)] \times d_t/\text{days}$**

SM cap eg: 1-yr cap, $1 m, 6-month SOFR cap 5%, reset 15 July (first period excluded — known at inception); SOFR 5.5% → paid next 15 Jan (184 days): 1,000,000 × 0.005 × 184/360 = **$2,555.56**.

⑩ ICAI patterns

- **Jan 2026 Q4(a)**: borrower **buys cap**, depositor **buys floor**. Lump-sum premium converted to per-period: **$P = A \times r_p \div \text{PVAF}(i, n)$** ($i$ = fixed rate per period). AB Ltd ₹5 m, 0.4%, PVAF(4%, 10) 8.1108 → **₹2,466**; gain 2% × ½ × 5 m = 50,000 → net **₹47,534**. XY Ltd ₹2 m, 0.5%, PVAF(4%, 6) 5.2421 → **₹1,908**; net **₹18,092**. ICAI also accepted $n - 1$ resets.
- **Cap to peg at 9%** (MTP): pay actual rate, receive excess over 9% → net cost never above 9%; years at or below 9% → option lapses. Cap preferred to a collar here — selling the floor at 9% causes outflow when rates fall below 9%.
- **May 2026 Q4(c)**: zero-cost collar — value of cap = value of floor (PV of expected payoffs) → solve the floor rate.

> ⚠ **TRAP**
> Cap / floor is per **reset period** — multiply by the fraction of the year (½ for 6-month resets). The first period is usually excluded (rate already known).

11 Interest rate swaps (SM 4.2.3)

- → Counterparties exchange payments indexed to two different rates on a **notional** principal that is **never exchanged**. Early swap dealers took a brokerage fee only; contract was between the end users
- → **Fixed / floating swap**: fixed rate · floating rate reset periodically · notional · term with reset and payment dates. Settlement **2 days after trade date**. Fixed rate quoted as **All-In-Cost (AIC)** against a flat floating index
- → Fixed payment (semi-annual) = $N \times AIC \times 180/360$ (30-day months, 360-day year); floating receipt = $N \times R \times d_t/360$

Type	SM meaning
Plain vanilla (generic / coupon)	Fixed ↔ floating on notional; floating = SOFR, €STER, SONIA, MIBOR, PLR
Basis rate (non-generic)	Both legs floating on different benchmarks (eg 1-month vs 3-month SOFR)
Asset swap	Fixed-rate investments (bonds) swapped for floating-rate investments (an index)
Amortising	Notional **declines** over the life — for issuers of redeemable bonds / debentures (hedges the redemption profile)

12 Swaptions (SM 4.2.4) — TYK Q2

- → Option on an IRS — right, not obligation, to enter a swap at a future date, fixed rate and term. **Payer (call) swaption**: pay fixed, receive floating. **Receiver (put) swaption**: receive fixed, pay floating
- → **Features**: option on a **forward-start IRS** with terms fixed at inception · "**3-month into 5-year**" = option to enter a 5-yr IRS in 3 months · option period = trade date to expiry · premium in **basis points** · can be **cash-settled** (MTM off the forward curve). Pricing: model of distribution of the forward zero-coupon curve
- → **Uses**: (a) traders & treasurers (b) speculate / hedge swap books (c) hedge **embedded optionality** (d) borrowers targeting an acceptable rate (e) businesses **tendering for contracts** (f) protection on **callable / puttable bonds**

13 Correlation — how Ch 12 actually connects

Ch 12 idea	The actual link (app-written cross-reference; the Ch 12 side is SM)
Price & reinvestment risk	**Ch 5**: same two risks that **immunization** offsets (duration = holding period); duration measures sensitivity of capital, gap measures sensitivity of earnings.
Cap, floor, collar	**Ch 9**: cap = series of calls, floor = series of puts on the rate; collar = long call + short put — same as a zero-cost option collar on shares.
FRA settlement discounted at RR	**Ch 10**: same idea as early settlement of a forward — money paid at the start of the period is worth less, so discount at the actual rate.
IRS / swaptions	**Ch 10**: currency swaps use the same comparative-advantage logic; **Ch 6**: IRS changes the duration of a bond portfolio (active fixed-income strategy).
IRF conversion factor, CTD	**Ch 5**: bond price = PV of coupons + redemption — CF makes bonds with different coupons / maturities comparable.
SOFR / SONIA / MIBOR	**Ch 11**: syndicated loans are priced at SOFR + spread; **FR Ind AS 109**: IBOR reform and hedge accounting.

Past papers

⑫ Interest Rate Risk Management — what ICAI asked (Sep 2025, Jan 2026, May 2026)

21 marks *set across the last three attempts (5.9% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	Q2(b)	4	FRA final settlement amount for a borrower, for a higher and a lower actual rate (formula with days/year basis) (Numerical)
Jan 2026	Q4(a)	6	Interest rate options: strategy for borrower (buy cap) and depositor (buy floor), premium from lump-sum premium rate, net gain/loss from hedging (Numerical)
May 2026	Q4(c) [or]	4	Zero-cost collar on a floating-rate loan: value of cap = value of floor (intrinsic value, PV of payoffs); solve the floor rate (Numerical)
May 2026	Q5(b)	7	Re-pricing gap (RSA - RSL): negative gap, favourable/unfavourable scenario, Earnings at Risk = gap x rate change x 1/4 versus 8% of NII limit, hedging decision (Numerical)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Write a short note on Forward Rate Agreements.

ICAI answer: "Please refer paragraph 4.1.2" – SM paragraph 4.1.2 reproduced

4.1.2 Forward Rate Agreements (FRAs)

A Forward Rate Agreement (FRA) is an agreement between two parties through which a borrower/ lender protects itself from the unfavourable changes to the interest rate. Unlike futures FRAs are not traded on an exchange thus are called OTC product. Following are main features of FRA.

- Normally it is used by banks to fix interest costs on anticipated future deposits or interest revenues on variable-rate loans indexed to Benchmark Interest Rate e.g. SOFR, ESTER, MIBOR etc.
- It is an off-Balance Sheet instrument.
- It does not involve any transfer of principal. The principal amount of the agreement is termed "notional" because, while it determines the amount of the payment, actual exchange of the

principal never takes place.

- It is settled at maturity in cash representing the profit or loss. A bank that sells an FRA agrees to pay the buyer the increased interest cost on some "notional" principal amount if Reference Rate of some specified maturity is above a stipulated "Forward Interest Rate" on the contract maturity or settlement date. Conversely, the buyer agrees to pay the seller any decrease in interest cost if Reference Rate fall below the forward rate.
- Final settlement of the amounts owed by the parties to an FRA is determined by the formula

$$= \frac{(N)(RR - FR)(dtm/DY)}{[1 + RR(dtm/DY)]} \times 100$$

Where,

N = the notional principal amount of the agreement;

RR = Reference Rate for the maturity specified by the contract prevailing on the contract settlement date; typically SOFR, ESTER or MIBOR

FR = Agreed-upon Forward Rate; and

dtm = maturity of the forward rate, specified in days (FRA Days)

DY = Day count basis applicable to money market transactions which could be 360 or 365 days.

If Reference Rate > FR then the seller owes the payment to the buyer, and if Reference Rate < FR then the buyer owes the seller the absolute value of the payment amount determined by the above formula.

- The differential amount is discounted at post change (actual) interest rate as it is settled in the beginning of the period not at the end.

Example

Suppose two banks enter into an agreement specifying:

- a forward rate of 5 percent on a Eurodollar deposit with a three-month maturity;
- a $1 million notional principal; and settlement in one month.

Such an agreement is termed a 1x4 FRA because it fixes the interest rate for a deposit to be placed after one month and maturing four months after the date the contract is negotiated.

If the three-month SOFR is 6 percent on the contract settlement date, the seller would owe the buyer the difference between 6 and 5 percent interest on $1 million for a period of 90 days.

Every 1 basis point change in the interest rate payable on a principal of $1 million for a 90-day maturity changes interest cost by $25, so that the increase in the interest cost on a three-month Eurodollar deposit over the specified forward rate in this case is $25 x 100 basis points = $2,500.

The $2,500 difference in interest costs calculated above is discounted back three months using the actual three-month SOFR prevailing on the settlement date.

Thus, if 90-day SOFR turns out to be 6 percent on the contract maturity date the buyer would receive $2,463.05 = $2,500/[1 + 0.06(90/360)].

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Thus, if 90-day SOFR turns out to be 6 percent on the contract maturity date the buyer would receive $2,463.05 = $2,500/[1 + 0.06(90/360)].

Theory Q2

2. What do you know about swaptions and their uses?

ICAI answer: "Please refer paragraph 4.2.4" – SM paragraph 4.2.4 reproduced

4.2.4 Swaptions

An interest rate swaption is simply an option on an interest rate swap. It gives the holder the right but not the obligation to enter into an interest rate swap at a specific date in the future, at a particular fixed rate and for a specified term.

There are two types of swaption contracts: -

- A **fixed rate payer swaption** (also called Call Swaption) gives the owner of the swaption the right but not the obligation to enter into a swap where they pay the fixed leg and receive the floating leg.
- A **fixed rate receiver swaption** (also called Put Swaption) gives the owner of the swaption the right but not the obligation to enter into a swap in which they will receive the fixed leg, and pay the floating leg.

4.2.4.1 Principal features of Swaptions

A. A swaption is effectively an option on a forward-start IRS, where exact terms such as the fixed rate of interest, the floating reference interest rate and the tenure of the IRS are established upon conclusion of the swaption contract.

B. A 3-month into 5-year swaption would therefore be seen as an option to enter into a 5-year IRS, 3 months from now.

C. The 'option period' refers to the time which elapses between the transaction date and the expiry date.

D. The swaption premium is expressed as basis points.

E. Swaptions can be cash-settled; therefore, at expiry they are marked to market off the applicable forward curve at that time and the difference is settled in cash.

4.2.4.2 Pricing of Swaptions

The pricing methodology depends upon setting up a model of probability distribution of the forward zero-coupon curve which undoes a Market process.

2.2.4.3 Uses of Swaptions

(a) Swaptions can be applied in a variety of ways for both active traders as well as for corporate treasurers.

(b) Swap traders can use them for speculation purposes or to hedge a portion of their swap books.

(c) Swaptions have become useful tools for hedging embedded optionality which is common to the natural course of many businesses.

(d) Swaptions are useful to borrowers targeting an acceptable borrowing rate.

(e) Swaptions are also useful to those businesses tendering for contracts.

(f) Swaptions also provide protection on callable/puttable bond issues.

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Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical QI

1. M/s. Parker & Co. is contemplating to borrow an amount of ₹60 crores for a Period of 3 months in the coming 6 month's time from now. The current rate of interest is 9% p.a., but it may go up in 6 month's time. The company wants to hedge itself against the likely increase in interest rate.

   The Company's Bankers quoted an FRA (Forward Rate Agreement) at 9.30% p.a.

   What will be the Final settlement amount, if the actual rate of interest after 6 months happens to be (i) 9.60% p.a. and (ii) 8.80% p.a.?

Solution — ICAI

1. Final settlement amount shall be computed by using formula:

$$= \frac{(N)(RR - FR)(dtm/DY)}{[1 + RR(dtm/DY)]}$$

Where,

N = the notional principal amount of the agreement;

RR = Reference Rate for the maturity specified by the contract prevailing on the contract settlement date;

FR = Agreed-upon Forward Rate; and

dtm = maturity of the forward rate, specified in days (FRA Days)

DY = Day count basis applicable to money market transactions which could be 360 or 365 days.

Accordingly,

If actual rate of interest after 6 months happens to be 9.60%

$$= \frac{(₹60\,crore)(0.096 - 0.093)(3/12)}{[1 + 0.096(3/12)]}$$

$$= \frac{(₹60\,crore)(0.00075)}{1.024} = ₹\ 4,39,453.13$$

Thus banker will pay Parker & Co. a sum of ₹ 4,39,453.13 or ₹ 4,39,453.

If actual rate of interest after 6 months happens to be 8.80%

$$= \frac{(₹60\,crore)(0.088 - 0.093)(3/12)}{[1 + 0.088(3/12)]}$$

$$= \frac{(₹60\,crore)(-0.00125)}{1.022} = -₹\ 7,33,855.19$$

Thus Parker & Co. will pay banker a sum of ₹ 7,33,855.19 or ₹ 7,33,855.

Note: Alternatively, students may solve the question on basis of different days count conventions instead of months (as considered in above calculations) i.e. 90 days/360 days, 90 days/ 365 days, 91 days/360 days or 91 days/365 days.

Practical Q2

2. TM Fincorp has bought a 6 x 9 ₹ 100 crore Forward Rate Agreement (FRA) at 5.25%. On fixing date reference rate i.e. MIBOR turns out be as follows:

Period	Rate (%)
3 months	5.50
6 months	5.70
9 months	5.85

You are required to determine:

(a) Profit/Loss to TM Fincorp. in terms of basis points.

(b) The settlement amount.

(Assume 360 days in a year)

Solution — ICAI

2. **(a)** TM will make a profit of 25 basis points since a 6X9 FRA is a contract on 3-month interest rate in 6 months, which turns out to be 5.50% (higher than FRA price).

(b) The settlement amount shall be calculated by using the following formula:

$$\frac{N(RR-FR)(dtm/360)}{1+RR(dtm/360)}$$

Where

N = Notional Principal Amount

RR = Reference Rate

FR = Agreed upon Forward Rate

Dtm = FRA period specified in days.

Accordingly:

$$\frac{100\,crore(5.50\%-5.25\%)(90/360)}{1+0.055(90/360)} = ₹\ 6,16,523$$

Hence there is profit of ₹ 6,16,523 to TM Fincorp.

Practical Q3

3. XYZ Limited borrows £ 15 Million of six months SONIA + 10.00% for a period of 24 months. The company anticipates a rise in SONIA, hence it proposes to buy a Cap Option from its Bankers at the strike rate of 8.00%. The lump sum premium is 1.00% for the entire reset periods and the fixed rate of interest is 7.00% per annum. The actual position of SONIA during the forthcoming reset period is as under:

Reset Period	SONIA
1	9.00%
2	9.50%
3	10.00%

You are required to show how far interest rate risk is hedged through Cap Option.

For calculation, work out figures at each stage up to three decimal points and amount nearest to £. It should be part of working notes.

Solution — ICAI

3. First of all we shall calculate premium payable to bank as follows:

$$P = \frac{rp}{\left[(1\div i) - \frac{1}{i\times(1+i)^t}\right]} \text{ X A or } \frac{rp}{PVAF(3.5\%,4)} \times A$$

Where

P = Premium

A = Principal Amount

rp = Rate of Premium

i = Fixed Rate of Interest

t = Time

$$= \frac{0.01}{\left[(1/0.035) - \frac{1}{0.035 \times 1.035^4}\right]} \times £15,000,000 \text{ or } \frac{0.01}{(0.966 + 0.933 + 0.901 + 0.871)}$$

$$\times £15,000,000$$

$$= \frac{0.01}{\left[(28.571) - \frac{1}{0.0402}\right]} \times £15,000,000 = £\ 40,595 \text{ or } \frac{£150,000}{3.671} = £\ 40,861$$

Now we see the net payment received from bank

Reset Period	Additional interest due to rise in interest rate	Amount received from bank	Premium paid to bank*	Net Amt. received from bank
1	£ 75,000	£ 75,000	£ 40,861	£34,139
2	£ 112,500	£ 112,500	£ 40,861	£71,639
3	£ 150,000	£ 150,000	£ 40,861	£109,139
TOTAL	£ 337,500	£ 337,500	£122,583	£ 214,917

Thus, from above it can be seen that interest rate risk amount of £ 337,500 reduced by £ 214,917 by using of Cap option.

* Alternatively, if premium paid is considered as £ 40,595, then above figure of £ 214,917 shall be changed to £ 215,715.

Practical Q4

4. Suppose a dealer quotes 'All-in-cost' for a generic swap at 8% against six month SOFR flat. If the notional principal amount of swap is US$ 5,00,000.

   (i) Calculate semi-annual fixed payment.

   (ii) Find the first floating rate payment for (i) above if the six month period from the effective date of swap to the settlement date comprises 181 days and that the corresponding SOFR was 6% on the effective date of swap.

   In (ii) above, if the settlement is on 'Net' basis, how much the fixed rate payer would pay to the floating rate payer?

   Generic swap is based on 30/360 days basis.

Solution — ICAI

4. (i) Semi-annual fixed payment

= (N) (AIC) (Period)

Where N = Notional Principal amount = US$ 5,00,000

AIC = All-in-cost = 8% = 0.08

$$= 5,00,000 \times 0.08 \left(\frac{180}{360}\right)$$

= 5,00,000 × 0.08 (0.5)

= 5,00,000 × 0.04 = ₹20,000/-

(ii) Floating Rate Payment

$$= N\,(SOFR) \left(\frac{dt}{360}\right)$$

$$= 5,00,000 \times 0.06 \times \frac{181}{360}$$

= 5,00,000 × 0.06 (0.503) or 5,00,000 × 0.06 (0.502777)

= 5,00,000 × 0.03018 or 5,00,000 × 0.030166 = US$ 15,090 or 15,083

(iii) Net Amount

= (i) – (ii)

= US$ 20,000 – US$ 15,090 = US$ 4,910

or = US$ 20,000 – US$ 15,083 = US$ 4,917

Practical QS

5. Derivative Bank entered into a plain vanilla swap through on OIS (Overnight Index Swap) on a principal of ₹ 10 crores and agreed to receive MIBOR overnight floating rate for a fixed payment on the principal. The swap was entered into on Monday, 2nd August, 2010 and was to commence on 3rd August, 2010 and run for a period of 7 days.

Respective MIBOR rates for Tuesday to Monday were:

7.75%,8.15%,8.12%,7.95%,7.98%,8.15%.

If Derivative Bank received ₹ 317 net on settlement, calculate Fixed rate and interest under both legs.

Notes:

(i) Sunday is Holiday.

(ii) Work in rounded rupees and avoid decimal working.

Solution — ICAI

5.

Day	Principal (₹)	MIBOR (%)	Interest (₹)
Tuesday	10,00,00,000	7.75	21,233
Wednesday	10,00,21,233	8.15	22,334
Thursday	10,00,43,567	8.12	22,256
Friday	10,00,65,823	7.95	21,795
Saturday & Sunday (*)	10,00,87,618	7.98	43,764
Monday	10,01,31,382	8.15	22,358
Total Interest @ Floating			1,53,740
Less: Net Received			317
Expected Interest @ fixed			1,53,423
Thus Fixed Rate of Interest			0.07999914
Approx.			8%

(*) i.e. interest for two days.

Note: Alternatively, answer can also be calculated on the basis of 360 days in a year.

Practical Q6

6. A Inc. and B Inc. intend to borrow $200,000 and $200,000 in ¥ respectively for a time horizon of one year. The prevalent interest rates are as follows:

Company	¥ Loan	$ Loan
A Inc	5%	9%
B Inc	8%	10%

The prevalent exchange rate is $1 = ¥120.

They entered in a currency swap under which it is agreed that B Inc will pay A Inc @ 1% over the ¥ Loan interest rate which the later will have to pay as a result of the agreed currency swap whereas A Inc will reimburse interest to B Inc only to the extent of 9%. Keeping the exchange rate invariant, quantify the opportunity gain or loss component of the ultimate outcome, resulting from the designed currency swap.

Solution — ICAI

6.

Opportunity gain of A Inc under currency swap	Receipt	Payment	Net
Interest to be remitted to B. Inc in $ 2,00,000 x 9% = $18,000 Converted into ($18,000 x ¥120)		¥21,60,000	
Interest to be received from B. Inc in $ converted into ¥ (6%x $2,00,000 x ¥120)	¥14,40,000	-	
Interest payable on ¥ loan	-	¥12,00,000	
	¥14,40,000	¥33,60,000	
Net Payment	¥19,20,000	-	
	¥33,60,000	¥33,60,000	
$ equivalent paid ¥19,20,000 x (1/¥120)			$16,000
Interest payable without swap in $			$18,000
Opportunity gain in $			$ 2,000

Opportunity gain of B inc under currency swap	Receipt	Payment	Net
Interest to be remitted to A. Inc in ($ 2,00,000 x 6%)		$12,000	
Interest to be received from A. Inc in ¥ converted into $ = ¥21,60,000/¥120	$18,000		
Interest payable on $ loan@10%	-	$20,000	
	$18,000	$32,000	
Net Payment	$14,000$32,000	- $32,000	
¥ equivalent paid $14,000 X ¥120			¥16,80,000
Interest payable without swap in ¥ ($2,00,000 X ¥ 120 X 8%)			¥19,20,000
Opportunity gain in ¥			¥ 2,40,000

Alternative Solution

Cash Flows of A Inc

(i) At the time of exchange of principal amount

Transactions		Cash Flows
Borrowings	$2,00,000 x ¥120	+ ¥240,00,000
Swap		- ¥240,00,000
Swap		+$2,00,000
Net Amount		+$2,00,000

(ii) At the time of exchange of interest amount

Transactions		Cash Flows
Interest to the lender	¥240,00,000X5%	¥12,00,000
Interest Receipt from B Inc.	¥2,00,000X120X6%	¥14,40,000
Net Saving (in $)	¥2,40,000/¥120	$2,000
Interest to B Inc.	$2,00,000X9%	-$18,000
Net Interest Cost		-$16,000

A Inc. used $2,00,000 at the net cost of borrowing of $16,000 i.e. 8%. If it had not opted for swap agreement the borrowing cost would have been 9%. Thus there is saving of 1%.

Cash Flows of B Inc

(i) At the time of exchange of principal amount

Transactions		Cash Flows
Borrowings		+ $2,00,000
Swap		- $2,00,000
Swap	$2,00,000X¥120	+¥240,00,000
Net Amount		+¥240,00,000

(ii) At the time of exchange of interest amount

Transactions		Cash Flows
Interest to the lender	$2,00,000X10%	- $20,000
Interest Receipt from A Inc.		+$18,000
Net Saving (in ¥)	-$2,000X¥120	- ¥2,40,000
Interest to A Inc.	$2,00,000X6%X¥120	- ¥14,40,000
Net Interest Cost		- ¥16,80,000

B Inc. used ¥240,00,000 at the net cost of borrowing of ¥16,80,000 i.e. 7%. If it had not opted for swap agreement the borrowing cost would have been 8%. Thus there is saving of 1%.

Practical Q7

7. A textile manufacturer has taken floating interest rate loan of ₹ 40,00,000 on 1st April, 2012. The rate of interest at the inception of loan is 8.5% p.a. interest is to be paid every year on 31st March. In the month of October 2012, the Central bank of the country releases following projections about the interest rates likely to prevail in future.

(i) On 31st March, 2013, at 8.75%; on 31st March, 2014 at 10% on 31st March, 2015 at 10.5% and on 31st March, 2016 at 7.75%. Show how the borrower can hedge the risk using Option Cap arising out of expected rise in the rate of interest when he wants to peg his interest cost at 8.50% p.a.

(ii) Assume that the premium negotiated by both the parties is 0.75% to be paid at once on 1st October, 2012 and the actual rate of interest on the respective due dates happens to be as: on 31st March, 2013 at 10.2%; on 31st March, 2014 at 11.5%; on 31st March, 2015 at 9.25%; on 31st March, 2016 at 8.25%. Show how the settlement will be executed on the perspective interest due dates.

Solution — ICAI

7. As borrower does not want to pay more than 8.5% p.a., on this loan where the rate of interest is likely to rise beyond this, hence, he has hedge the risk by entering into an agreement to buy interest rate caps with the following parameters:

- Notional Principal: ₹ 40,00,000/-
- Strike rate: 8.5% p.a.
- Reference rate: the rate of interest applicable to this loan
- Calculation and settlement date: 31st March every year
- Duration of the caps: till 31st March 2016
- Premium for caps: negotiable between both the parties

To purchase the caps this borrower is required to pay the premium upfront at the time of buying caps. The payment of such premium will entitle him with right to receive the compensation from the seller of the caps as soon as the rate of interest on this loan rises above 8.5%. The compensation will be at the rate of the difference between the rate of none of the cases the cost of this loan will rise above 8.5% calculated on ₹ 40,00,000/-. This implies that in none of the cases the cost of this loan will rise above 8.5%. This hedging benefit is received at the respective interest due dates at the cost of premium to be paid only once.

The premium to be paid on 1st October 2012 is **30,000/-** (₹ 40,00,000 x 0.75/100). The payment of this premium will entitle the buyer of the caps to receive the compensation from the seller of the caps whereas the buyer will not have obligation. The compensation received by the buyer of caps will be as follows:

On 31st March 2013

The buyer of the caps will receive the compensation at the rate of 1.70% (10.20 - 8.50) to be calculated on ₹ 40,00,000, the amount of compensation will be ₹ 68000/- (40,00,000 x 1.70/100).

On 31st March 2014

The buyer of the caps will receive the compensation at the rate of 3.00% (11.50 – 8.50) to be calculated on ₹ 40,00,000/-, the amount of compensation will be ₹ 120000/- (40,00,000 x 3.00/100).

On 31st March 2015

The buyer of the caps will receive the compensation at the rate of 0.75% (9.25 – 8.50) to be calculated on ₹ 40,00,000/-, the amount of compensation will be ₹ 30,000 (40,00,000 x 0.75/100).

On 31st March 2016

The buyer of the caps will not receive the compensation as the actual rate of interest is 8.25% whereas strike rate of caps is 8.5%. Hence, his interest liability shall not exceed 8.50%.

Thus, by paying once the premium upfront buyer of the caps gets the compensation on the respective interest due dates without any obligations.

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